Entity resolution and data QC

Veridion POC Simulation, Part 1. All work is in `presales_work_final.xlsx`; start with its `README` tab, then `Reasoning`.

The task. A client sent 592 company names. The matching engine returned up to 5 candidate records for each (2,951 rows). I had to pick the right candidate per company, then check the quality of the chosen data and propose how to curate it for a manufacturing client doing procurement spend analysis.

592	252	273	67
client companies	Matched	Review (provisional)	No match

19
records shared by 2-3 inputs

How I thought about it

1. **Compare each candidate with its own input, then choose one winner**

   Every candidate gets a score out of 100: name 40, country 20, street 20, city 10, postal code 10. Fields the client did not give are not counted against the candidate. The weights are my judgement, not calculated from the data, and the formulas are live in `Scored_candidates` so they can be changed.

2. **The name matters most**

   Names are cleaned first (case, legal suffixes). A name also counts if it equals one of the candidate's legal or commercial names. A match on the first word only earns fewer points.

3. **Ties are not decided by chance**

   If two candidates score the same, the one with more filled fields wins. If that is equal too, the row is flagged and I decided by hand from company registers and websites: 7 cases, 2 matches and 5 "no match" (the real company was not among the candidates).

4 **I tested the automatic rule instead of trusting it**

I checked a sample of 30 accepted rows, 10 for each way of matching. The result changed the rule: **Auto-accept now needs the full name to match.**

5 **I never delete rows**

`Final_matches` has exactly one row per client company, with a status, a confidence and a recommended action. Where nothing reliable exists the row says "No match" and keeps the closest candidate only as a hint.

6 **Duplicates**

19 company records were chosen for 2-3 client rows. 4 are the same company written twice in the client list; 15 are different entities of one group matched to a single group record. The client decides whether a group-level record is enough.

What the sample showed

Way of matching	Rows in file	Right	Group-level	Unclear	Wrong
Exact main name	161	9	1	0	0
Legal or commercial name	90	9	1	0	0
First word only	152	3	3	1	3

10 rows checked per line. "Group-level" means the right company group but not the exact legal entity. First-word matches were right and wrong at the same score (90-92%), so the score could not separate them. Final split: 251 Auto-accept, 279 Review, 62 Unmatched.

Data QC and curation proposal

I wrote 11 consistency rules that compare fields inside one record, and ran them on the Matched and Review rows. A flag means "look here", not "this is wrong". Share of rows flagged:

Check	Matched (of 252)	Review (of 273)
Website differs from email domain	20%	21%

Employee count looks like a modelled round value (75, 150, 15...)	18%	12%
No website, email or phone	6%	13%
Revenue per employee implausible	2%	6%
Website unrelated to the company name	3%	4%
Industry and sector contradict each other	4%	4%

Records with no flag at all: 131 of 252 Matched, 119 of 273 Review. Review rows are also emptier on most fields, so treat them as leads until the match is confirmed.

Proposal

- Fix only what is safe: clean names, blank impossible years.
- Suggest + check the email domain as the website. For groups the email is often the parent company, so it is not a safe automatic change.
- Verify wrong websites, contradicting industries and impossible revenue per employee.
- Show as estimate modelled employee and revenue figures, regional phones and old records.

For a procurement client the fields that matter most are industry, size, country and contact, so those flags come first.

What is not done

- Review (279) and Unmatched (62) rows were not checked by hand. Next: 10 rows per Review group on the web, with accuracy per group.
- Some sample verdicts rest on the data, not on the web. I would re-check 5 of them on the web.
- The QC thresholds and the industry pairs are my own choices. The counts show where to look, not how many errors exist. I would also test how many picks change if the scoring weights move.

> - Flagged rows were not verified on the web.

AI use

I used Claude to build the first version of the scoring formulas, to run the checks on the data and to draft the wording of the tabs and this page. I used Microsoft Copilot to reorder the input columns (step 1). I defined the approach and the rules, decided the ties, the sample verdicts and the rule change, verified part of the sample on the web, and corrected the wording. Some web checks were limited by the tools Claude had, so those verdicts rest on the data and are marked as such in `Sample_check`.